MT1011 ENGINEERING MATERIALS
STRUCTURE · CRYSTALLOGRAPHY · DEFECTS

Name: .....
ID: .....

ATOMS & BONDING

Atom: atomic number Z = protons; neutral electrons $= Z$; ion electrons $= Z -$ charge. Mass number $A_m = Z + N$ (neutrons N); atomic weight M is molar mass, not A_m. **Electronic structure:** orbitals s^2, p^6, d^{10}. Fill $4s$ before $3d$; remove $4s$ electrons first when forming transition-metal cations. **Ionic:** electron transfer/electrostatic attraction; **covalent:** shared electron pair/directional; **metallic:** positive cores + delocalised electrons; **secondary:** van der Waals/dipoles/H bonds (usually weaker). Ceramics: ionic and/or covalent.

$$\%IC = 100\left[1 - e^{-\frac{1}{4}(\chi_A - \chi_B)^2}\right]$$

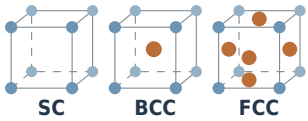
χ_A, χ_B: Pauling electronegativities. Larger $|\Delta\chi| \Rightarrow$ more ionic; smallest $|\Delta\chi| \Rightarrow$ most covalent. Paper data: $\chi_{Cu} = 1.9$, F 4.0, Cl 3.0, S 2.5, O 3.5.

METHOD Bond character: find $|\chi_A - \chi_B| \rightarrow$ square $\rightarrow$ substitute in exponential $\rightarrow \times 100$.

$$F_N(r) = F_A(r) + F_R(r), \quad F_N(r_0) = 0$$

F_A attraction, F_R repulsion (signed), r_0 equilibrium separation. Energy minimum at r_0; deeper well = higher bond energy, steeper near-minimum curvature = greater modulus; asymmetric well = thermal expansion.

CUBIC / HEXAGONAL CELLS



	SC	BCC	FCC	HCP
Atoms n	1	2	4	6
Coord. no.	6	8	12	12
APF	.52	.68	.74	.74
Touch along	edge	body diag.	face diag.	close-pack
a/R	2	$4/\sqrt{3}$	$2\sqrt{2}$	-

Count atom shares: corner $\frac{1}{8}$, edge $\frac{1}{4}$, face $\frac{1}{2}$, body 1. R atomic radius; a cubic edge.

$$\underbrace{\sqrt{3}a = 4R}_{\text{BCC}} \qquad \underbrace{\sqrt{2}a = 4R}_{\text{FCC}}$$



HCP prism
$ABAB$ layers
Ideal $c/a = 1.633$

FCC layers: $ABCABC$; HCP: $ABAB$. Both close-packed, coordination 12.

CRYSTAL / UNIT CELL

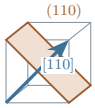
Crystalline = ordered periodic arrangement; amorphous = no long-range order. Unit cell = repeating unit; parameters a, b, c lengths and α, β, γ angles. Polycrystal = differently oriented grains.

$$\rho = \frac{nM}{N_A V_c}, \quad \text{APF} = \frac{n\frac{4}{3}\pi R^3}{V_c}$$

ρ theoretical mass density; M molar mass; $N_A = 6.022 \times 10^{23}$ mol^{-1}; $V_c = a^3$ for cubic; n atoms/cell. Use a in cm for ρ in g cm^{-3}.

DIRECTIONS · PLANES · FAMILIES

$[uvw]$ one **direction**, vector components
$\langle uvw \rangle$ symmetry-related **direction family**
(hkl) one **plane**, intercept reciprocals
$\{hkl\}$ symmetry-related **plane family**



Cubic examples: $\langle 100 \rangle$: axes (6 directions); $\langle 110 \rangle$: face diagonals (12), e.g. $[110], [1\bar{1}0], [101]$; $\langle 111 \rangle$: body diagonals (8). $\{100\}$ = cube faces; $\{110\}$ includes $(110), (1\bar{1}0), (101)$; $\{111\}$ includes $(1\bar{1}1)$. Families allow permutations and signs.

METHOD Direction: head − tail in fractional cell coordinates $\rightarrow$ clear fractions $\rightarrow$ reduce integers $\rightarrow$ bar negative components.

METHOD Plane: axis intercepts in a, b, c units $\rightarrow$ reciprocals $\rightarrow$ clear fractions $\rightarrow$ bars on negatives. Translate if plane passes through origin. Parallel axis $\Rightarrow \infty$ intercept $\Rightarrow 0$ index.

$$\frac{hx}{a} + \frac{ky}{b} + \frac{lz}{c} = 1, \quad d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

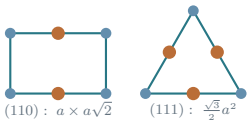
Intercept-form plane when origin chosen suitably; x, y, z Cartesian positions. d_{hkl} perpendicular plane spacing for **cubic** lattice only.

WATCH Plane $(hkl) \neq$ direction $[hkl]$. Translate vector freely; use atom *centres* to test plane membership.

LINEAR & PLANAR DENSITY

$$\rho_L = \frac{N_{\text{line}}}{L}, \qquad \rho_P = \frac{N_{\text{plane}}}{A}$$

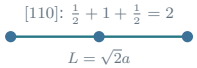
N_{line} effective atoms on chosen repeat segment, L segment length; N_{plane} effective atoms with centres on plane, A planar repeat area. Units atoms/nm and atoms/nm^2.



FCC plane counts: (100) square: 4 corners $\times \frac{1}{4} + 1 = 2$; (110) rectangle: 4 corners $\times \frac{1}{4} + 2$ edge atoms $\times \frac{1}{2} = 2$; (111) triangle: 3 vertices $\times \frac{1}{6} + 3$ edge midpoints $\times \frac{1}{2} = 2$.

$$\rho_P^{\text{FCC}}(100) = \frac{2}{a^2}$$
$$\rho_P^{\text{FCC}}(110) = \frac{2}{\sqrt{2}a^2}$$
$$\rho_P^{\text{FCC}}(111) = \frac{4}{\sqrt{3}a^2}$$

FCC densest: $\{111\}$ planes; $\langle 110 \rangle$ directions. **BCC dense direction:** $\langle 111 \rangle$.



$$\rho_L^{\text{FCC}}[100] = \frac{1}{a}, \qquad \rho_L^{\text{FCC}}[110] = \frac{2}{\sqrt{2}a}$$
$$\rho_L^{\text{FCC}}[111] = \frac{1}{\sqrt{3}a}, \quad \rho_L^{\text{BCC}}[111] = \frac{2}{\sqrt{3}a} = \frac{1}{2R}$$

METHOD Find structure $\rightarrow$ contact diagonal $\rightarrow$ $a(R)$ $\rightarrow$ choose *repeat* line/plane $\rightarrow$ count shared atoms $\rightarrow$ length/area $\rightarrow$ divide; check units.

7 SYSTEMS / 14 BRAVAIS

System	Cell geometry	Lattices
Cubic	$a = b = c$; all 90°	P,I,F
Tetragonal	$a = b \neq c$; all 90°	P,I
Orthorhombic	$a \neq b \neq c$; all 90°	P,C,I,F
Hexagonal	$a = b \neq c$; $\gamma = 120^\circ$	P
Rhombohedral	$a = b = c$; equal, non-90° angles	R
Monoclinic	one non-90° angle	P,C
Triclinic	unequal edges/oblique angles	P

= primitive, I body-centred, F face-centred, C base-centred, R rhombohedral. Crystal lattice + basis = crystal structure.

CRYSTAL DEFECTS



0D point: vacancy (missing host atom); substitutional solute (replaces host); interstitial solute (small atom in gap, e.g. C/N/O in Fe); self-interstitial (extra host atom). **1D line:** edge, screw, mixed dislocations. **2D surface:** grain, twin, phase boundaries and free surface.

$$N_v = N \exp\left(-\frac{Q_v}{k_B T}\right)$$

N_v equilibrium vacancies; N atomic lattice sites; Q_v energy per vacancy; T in kelvin; $k_B = 1.381 \times 10^{-23}$ J/K or 8.617×10^{-5} eV/K (match Q_v units).

Dislocations: edge = extra half-plane, Burgers vector **b** $\perp$ dislocation line; screw = shear spiral, **b** $\parallel$ line; mixed = combination. Dislocations glide on slip systems; motion enables plastic deformation. **Slip system:** slip *plane* + slip *direction*. FCC example $\{111\}\langle 110 \rangle$ (12). Grain boundary blocks dislocation transfer across misoriented grains.

STRENGTHENING QUICK RULES

Principle: harder dislocation motion $\Rightarrow$ higher yield strength/hardness; ductility often decreases.

$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

Hall–Petch: σ_y yield stress; σ_0 lattice friction stress; k_y material constant; d average grain size. Finer grains $\Rightarrow$ stronger (usual grain scale).

$$\%CW = 100\frac{A_0 - A_d}{A_0}$$

A_0, A_d: original and deformed cross-sectional areas. Cold-work amount $\uparrow \Rightarrow$ dislocation density/strength $\uparrow$, ductility $\downarrow$. Annealing can remove cold-work effects.

Solid-solution strengthening: size-mismatched solutes create strain fields and obstruct dislocations; pure metal cannot use this mechanism without added solute. Small solutes cause local tensile strain, large local compressive strain. **Refinement:** more grain boundaries obstruct dislocations.

WATCH Do not confuse lower grain size with lower grain-boundary area: smaller d usually *increases* boundary area.

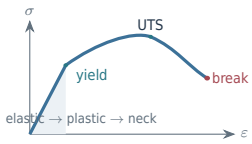
MATERIAL CLASSES

Metals: conducting, generally ductile. Ceramics: stiff/hard, brittle, strong in compression. Polymers: long chains, typically lightweight/low stiffness. Composites: two or more constituents, e.g. reinforcement + matrix.

MT1011 ENGINEERING MATERIALS
MECHANICS · STRENGTHENING · PHASE DIAGRAMS

Name: .....
ID: .....

TENSILE RESPONSE



Elastic: reversible strain; **plastic:** permanent strain. Initial slope E = Young modulus (stiffness). Yield σ_y = onset of plasticity; if no clear yield, use 0.2% offset (strain $= 0.002$). UTS = peak *engineering* stress; necking then fracture. Lower yield point and plateau may occur in mild steels.

$$\sigma = \frac{F}{A_0}, \qquad \varepsilon = \frac{L - L_0}{L_0}$$

F axial force; A_0 original area; L_0, L initial/current gauge lengths. Stress Pa (MPa if N/mm^2); strain dimensionless.

$$E = \frac{\Delta\sigma}{\Delta\varepsilon}, \qquad \sigma = E\varepsilon \text{ (linear elastic)}$$

E slope on initial straight segment; σ_y proof/yield stress; $\sigma_{\text{UTS}} = F_{\max}/A_0$.

$$\sigma_{\text{true}} = \frac{F}{A_{\text{inst}}}, \qquad \varepsilon_{\text{true}} = \ln(1+\varepsilon)$$

A_{inst} instantaneous area. Before necking, assuming uniform constant-volume deformation, $\sigma_{\text{true}} \approx \sigma(1+\varepsilon)$; *not* generally valid after necking.

$$\%EL = 100\frac{L_f - L_0}{L_0}$$
$$\%RA = 100\frac{A_0 - A_f}{A_0}$$

L_f final length; A_f final minimum area. Ductility measured by elongation or area reduction. Brittle often $\varepsilon_f \lesssim 5\%$.

$$U_r = \int_0^{\varepsilon_y} \sigma\, d\varepsilon = \frac{\sigma_y^2}{2E}$$

U_r elastic resilience / unit volume; ε_y elastic strain at yield, valid for linear elastic. Toughness = *total* area under stress–strain curve to fracture.

STRENGTH / TOUGHNESS / TESTS

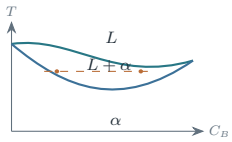
Stiffness $E \neq$ **strength** σ_y or UTS $\neq$ **toughness** (energy/volume) $\neq$ **hardness** (local indentation resistance). **Brittle ceramics:** low plasticity, often higher compressive than tensile strength. **MOR:** flexural strength from bending test. **Impact:** Charpy/Izod notched pendulum tests; energy absorbed $\approx mg(h - h')$, where m mass, g gravity, h, h' starting/ending heights. Charpy supported at both ends; Izod cantilever clamped. **Hardness:** Brinell ball, Vickers diamond pyramid, Rockwell penetration depth. Compare values only on compatible scales.

METHOD Stress questions: decide original vs current area → force/area; slope only in elastic region; use peak *force* for UTS; fracture for $\%EL/\%RA$.

SOLUTIONS / PHASE LANGUAGE

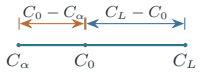
Component: element/compound of alloy. **Phase:** homogeneous composition/structure; single-phase vs two-phase fields. **Solubility limit:** maximum dissolved amount at a stated T. **Microstructure:** shape, size, proportion/distribution of phases. **Isomorphous:** complete solid miscibility (Cu–Ni, FCC). **Substitutional:** solute occupies host site; **interstitial:** occupies gap. **Hume–Rothery:** similar radius ($\lesssim 15\%$ difference), same crystal type, similar electronegativity, same valence favour extensive substitutional solubility. **Lines:** liquidus = liquid-only boundary (first solid on cooling); solidus = last liquid disappears; solvus = solid solubility boundary. Diagram usually equilibrium at stated pressure.

ISOMORPHOUS PHASES (Cu-Ni)



Top L, lens $L + \alpha$, below α solid solution. At given composition C_0 cooling crosses *liquidus* → first α then *solidus* → all α. Horizontal tie line endpoints give C_α, C_L.

TIE LINE + LEVER RULE



$$W_\alpha = \frac{C_L - C_0}{C_L - C_\alpha}$$
$$W_L = \frac{C_0 - C_\alpha}{C_L - C_\alpha}$$

C_0 overall composition; C_α, C_L solid/liquid compositions at that T; W_α, W_L *mass fractions*, summing to 1. For general left/right phases, replace α, L with corresponding labels.

METHOD 1. Read the *bottom-axis unit* (wt% vs at%). 2. Mark (C_0, T). 3. Name phase field. 4. Draw horizontal tie line. 5. Read endpoints at boundaries. 6. Use *opposite* lever arm. 7. Verify $0 \le W_i \le 1$ and $\sum W = 1$.

WATCH A phase's *composition* is a tie-line endpoint; its *fraction* is from the lever rule. Do not interchange them.

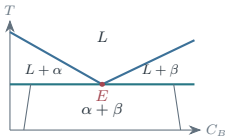
PHASE FRACTIONS vs STRUCTURES

Phase fractions W_α, W_β count *all* α, β (both primary and eutectic). **Microconstituents** distinguish *primary solid* and *eutectic mixture*. Eutectic fraction equals just-above-T_E liquid mass fraction, under equilibrium cooling.

$$W_{\text{eutectic}} = W_L(T_E^+)$$

Primary fraction $W_{\text{primary}} = 1 - W_{\text{eutectic}}$ when applicable; total W_α and W_β generally differ from these.

EUTECTIC TRANSFORMATIONS



$$L(C_E) \xrightarrow[\text{cooling}]{T_E} \alpha(C_{\alpha E}) + \beta(C_{\beta E})$$

T_E, C_E: eutectic temperature/composition; $C_{\alpha E}, C_{\beta E}$ end-point solid compositions at T_E. Left side $C_0 < C_E$: hypo-eutectic → primary α+ eutectic; right $C_0 > C_E$: hyper-eutectic → primary β+ eutectic.

Cool from liquid: at liquidus → first *primary* solid; in $L + \alpha$ or $L + \beta$, liquid composition follows liquidus; at T_E remaining liquid → $\alpha + \beta$; final structure = primary + fine eutectic mixture. Exactly C_E: all-eutectic microstructure. **Solvus:** below eutectic, falling solubility may precipitate second solid within former solid solution.

KEY LECTURE PHASE VALUES

Alloy / composition axis	T_E	C_E
Cu-Ag / wt% Ag	779°C	71.9
Pb–Sn / wt% Sn	183°C	61.9
Au–Ge / wt% Ge*	361°C	12.5

Cu–Ag: $C_{\alpha E} = 8.0$ wt% Ag (Cu-rich FCC), $C_{\beta E} = 91.2$ wt% Ag (Ag-rich FCC). Pb–Sn: $C_{\alpha E} = 18.3$ wt% Sn, $C_{\beta E} = 97.8$ wt% Sn.

Au–Ge (past paper): pure Au 1064.43°C; pure Ge 938.3°C. Read curved liquidus from supplied graph.

EUTECTIC COMPUTATION ROUTE

METHOD For first solid: draw vertical through C_0 until it meets liquidus; note T and solid-rich side. For last liquid: follow to eutectic horizontal T_E if within eutectic solidification range.

METHOD Liquid composition at T: horizontal line to *liquidus*, read C_L. Liquid fraction: tie-line endpoints C_L, C_S (pure rich-side solid $C_S \approx 100$ only if graph warrants); opposite-arm ratio.

$$W_L = \frac{C_S - C_0}{C_S - C_L}, \qquad W_S = 1 - W_L$$

C_S composition of solid at T; C_L liquid, C_0 bulk; these apply when $C_L < C_0 < C_S$ (Ge-rich L+solid case).

EXAM CHOICE CHECKS

Bonding: ionic% uses $(\Delta\chi)^2$; covalent ranking uses $|\Delta\chi|$. **Index:** $[uvw]$ head−tail; (hkl) intercept reciprocals; $\langle\rangle/\{\}$ families. **Density:** count shared centres, select matching geometric length/area, convert a–R. **Phase:** identify correct graph/axis, field, tie line, end compositions, then fraction. **Strengthening:** pure metals use grain refinement or work hardening; solute strengthening needs added atoms.

WATCH Always read new alloy data from the supplied diagram. Solid-solution strengthening requires foreign solute atoms.

PERIODIC TABLE atomic number + symbol

1 H																	2 He
3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
55 Cs	56 Ba	57-71 Ln	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra	89-103 An	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110 Ds	111 Rg	112 Cn	113 Nh	114 Fl	115 Mc	116 Lv	117 Ts	118 Og
			57 La	58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
			89 Ac	90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr